



Mata Kuliah : Pemrograman Mobile
Program Studi : D4 – Teknik Informatika
Semester : 5

Kelas : 2A TI
NIM : 244107020112
Nama : Muhammad Shabran
Pertemuan : 3

LAPORAN PRAKTIKUM

Aplikasi multi-page dengan GoRouter

Langkah	Keterangan
1.	Membuat Project dan Menambahkan Dependency <pre>PS C:\Users\muham\Documents\Tugas Kuliah\Semester 5\Mobile\244107020112_mobile_course> flutter create week3_navigation Creating project week3_navigation... Resolving dependencies in `week3_navigation`... (2.5s) Downloading packages... Got dependencies in `week3_navigation`. Wrote 131 files. All done! You can find general documentation for Flutter at: https://docs.flutter.dev/ Detailed API documentation is available at: https://api.flutter.dev/ If you prefer video documentation, consider: https://www.youtube.com/c/flutterdev In order to run your application, type: \$ cd week3_navigation \$ flutter run Your application code is in week3_navigation\lib\main.dart.</pre> <pre>PS C:\Users\muham\Documents\Tugas Kuliah\Semester 5\Mobile\244107020112_mobile_course> cd week3_navigation PS C:\Users\muham\Documents\Tugas Kuliah\Semester 5\Mobile\244107020112_mobile_course\week3_navigation> PS C:\Users\muham\Documents\Tugas Kuliah\Semester 5\Mobile\244107020112_mobile_course\week3_navigation> flutter pub add go_router Resolving dependencies... Downloading packages... (1:16.2s) + cupertino_ui 1.1.1 + flutter_localizations 0.0.0 from sdk flutter + flutter_web_plugins 0.0.0 from sdk flutter + go_router 18.0.2 + intl 0.20.3 + logging 1.3.0 + material_color_utilities 0.13.0 (0.13.1 available) + material_ui 1.5.0 + meta 1.18.3 (1.19.0 available) + test_api 0.7.12 (0.7.14 available) + vector_math 2.4.0 (2.4.3 available) Changed 7 dependencies! 4 packages have newer versions incompatible with dependency constraints. Try `flutter pub outdated` for more information.</pre>
2.	Mendefinisikan Router



	<pre>import 'package:flutter/material.dart'; import 'package:go_router/go_router.dart'; import 'pages/detail_page.dart'; import 'pages/home_page.dart'; void main() => runApp(const MyApp()); final _router = GoRouter(initialLocation: '/', routes: [GoRoute(path: '/', builder: (context, state) => const HomePage(), routes: [GoRoute(path: 'detail/:id', builder: (context, state) => DetailPage(id: state.pathParameters['id']!,),),],),],); class MyApp extends StatelessWidget { const MyApp({super.key}); @override Widget build(BuildContext context) { return MaterialApp.router(title: 'Week 3 - Navigation', routerConfig: _router, theme: ThemeData(colorsSchemeSeed: Colors.indigo, useMaterial3: true),); } }</pre>
3.	Membuat Halaman Home <pre>import 'package:flutter/material.dart'; import 'package:go_router/go_router.dart'; class HomePage extends StatelessWidget { const HomePage({super.key}); @override Widget build(BuildContext context) { return Scaffold(appBar: AppBar(title: const Text('Home')), body: ListView.builder(itemCount: 10, itemBuilder: (context, index) => ListTile(title: Text('Item \${index + 1}'), onTap: () => context.go('/detail/\${index + 1}'),),),); } }</pre>
4.	Membuat Halaman Detail

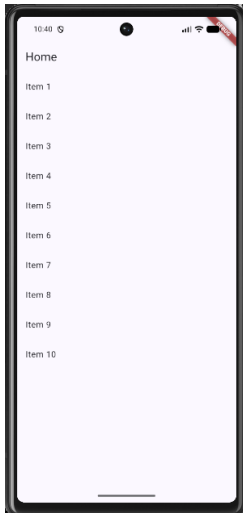


```
import 'package:flutter/material.dart';

class DetailPage extends StatelessWidget {
  final String id;
  const DetailPage({super.key, required this.id});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Detail $id')),
      body: Center(child: Text('Anda membuka item dengan id: $id')),
    );
  }
}
```

5. Hasil AntarMuka:
Home page :



Detail Page :

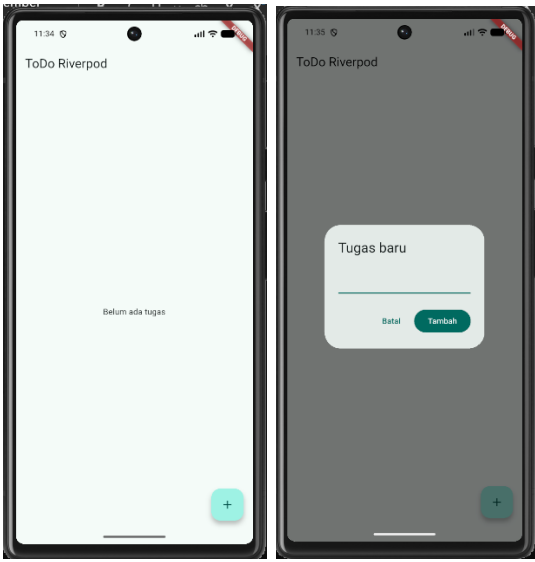




Praktikum: State management dengan Riverpod

Langkah	Keterangan
1.	Tambahkan Dependency Riverpod <pre>PS C:\Users\muham\Documents\Tugas Kuliah\Semester 5\Mobile\244107020112_mobile_course\week3_navigation> flutter pub add flutter_riverpod Resolving dependencies... Downloading packages... (13.9s) + crypto 3.0.7 + fixnum 1.1.1 + flutter_riverpod 3.4.3 + listen 1.0.1 + material_color_utilities 0.13.0 (0.13.1 available) + meta 1.18.3 (1.19.0 available) + riverpod 3.4.3 + state_notifier 1.0.0 + test_api 0.7.12 (0.7.14 available) + typed_data 1.4.0 + uuid 4.6.0 + vector_math 2.4.0 (2.4.3 available) Changed 8 dependencies! 4 packages have newer versions incompatible with dependency constraints. Try "flutter pub outdated" for more information.</pre>
2.	Membuat File ToDo <pre>import 'package:flutter/material.dart'; import 'package:flutter_riverpod/flutter_riverpod.dart'; class Todo { Todo(this.title, {this.done = false}); final String title; final bool done; Todo copyWith({String? title, bool? done}) => Todo(title ?? this.title, done: done ?? this.done); } class TodoListNotifier extends Notifier<List<Todo>> { @override List<Todo> build() => const []; void add(String title) => state = [...state, Todo(title)]; void toggle(int index) { final todos = [...state]; todos[index] = todos[index].copyWith(done: !todos[index].done); state = todos; } void remove(int index) => state = [...state]..removeAt(index); } final todoListProvider = NotifierProvider<TodoListNotifier, List<Todo>>(TodoListNotifier.new); class TodoPage extends ConsumerWidget { const TodoPage({super.key}); @override Widget build(BuildContext context, WidgetRef ref) { final todos = ref.watch(todoListProvider); return Scaffold(appBar: AppBar(title: const Text('ToDo Riverpod')), body: todos.isEmpty ? const Center(child: Text('Belum ada tugas')) : ListView.builder(itemCount: todos.length, itemBuilder: (context, index) => ListTile(leading: Checkbox(checked: todos[index].done, onChanged: () => ref.read(todoListProvider.notifier).toggle(index),), title: Text(todos[index].title),),), floatingActionButton: FloatingActionButton(onPressed: () => _showAddDialog(context, ref), child: const Icon(Icons.add),),); } void _showAddDialog(BuildContext context, WidgetRef ref) { final controller = TextEditingController(); showDialog(context: context, builder: (context) => AlertDialog(title: const Text('Tugas baru'), content: TextField(controller: controller, autofocus: true), actions: [TextButton(onPressed: () => Navigator.pop(context), child: const Text('Batal'),), FilledButton(onPressed: () { if (controller.text.trim().isNotEmpty) { ref .read(todoListProvider.notifier) .add(controller.text.trim()); Navigator.pop(context); } }, child: const Text('Tambah'),),],),); } }</pre>
3.	Mengonfigurasi main.dart



	<pre>import 'package:flutter/material.dart'; import 'package:flutter_riverpod/flutter_riverpod.dart'; import 'pages/todo_page.dart'; void main() => runApp(const ProviderScope(child: MyApp())); class MyApp extends StatelessWidget { const MyApp({super.key}); @override Widget build(BuildContext context) { return MaterialApp(title: 'Week 3 - ToDo', theme: ThemeData(colorSchemeSeed: Colors.teal, useMaterial3: true), home: const TodoPage(),); } }</pre>
4.	Hasil antarmuka: 

Praktikum : AsyncValue: loading, error, success

Langkah	Keterangan
1.	Salin kode ke project todo



```
import 'package:flutter/material.dart';
import 'package:flutter_riverpod/flutter_riverpod.dart';

class ProductsNotifier extends AsyncNotifier<List<String>> {
  @override
  Future<List<String>> build() async {
    await Future.delayed(const Duration(seconds: 2)); // simulasi network
    return ['Keyboard', 'Mouse', 'Monitor'];
  }

  Future<void> refresh() async {
    state = const AsyncLoading();
    state = await AsyncValue.guard(() => _fetch());
  }

  Future<List<String>> _fetch() async {
    await Future.delayed(const Duration(seconds: 1));
    return ['Keyboard', 'Mouse', 'Monitor', 'Headset'];
  }
}

final productsProvider =
  AsyncNotifierProvider<ProductsNotifier, List<String>>(
    ProductsNotifier.new);

class ProductPage extends ConsumerWidget {
  const ProductPage({super.key});

  @override
  Widget build(BuildContext context, WidgetRef ref) {
    final productsAsync = ref.watch(productsProvider);

    return Scaffold(
      appBar: AppBar(title: const Text('Produk')),
      body: productsAsync.when(
        loading: () => const Center(child: CircularProgressIndicator()),
        error: (err, stack) => Center(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.min,
            children: [
              Text('Gagal memuat: $err'),
              FilledButton(
                onPressed: () => ref.invalidate(productsProvider),
                child: const Text('Coba lagi'),
              ),
            ],
          ),
        ),
        data: (products) => ListView.builder(
          itemCount: products.length,
          itemBuilder: (context, index) =>
            ListTile(title: Text(products[index])),
        ),
      ),
    );
  }
}
```

```
import 'package:flutter/material.dart';
import 'package:flutter_riverpod/flutter_riverpod.dart';
import 'pages/todo_page.dart'; Tab to Jump

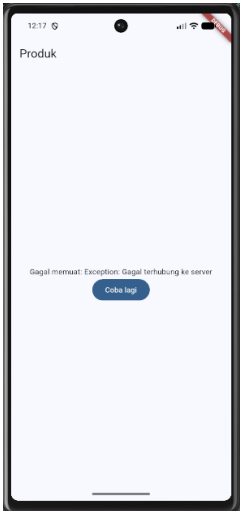
void main() => runApp(const ProviderScope(child: MyApp()));

class MyApp extends StatelessWidget {
  const MyApp({super.key});

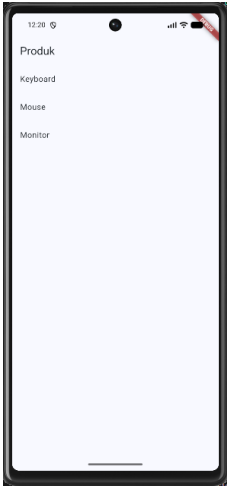
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Week 3 - Async',
      theme: ThemeData(colorSchemeSeed: Colors.blue, useMaterial3: true),
      home: const ProductPage(),
    );
  }
}
```

Amati tampilan 2 detik pertama:



	
2.	Mengubah build(): <pre>Future<List<String>> build() async { await Future.delayed(const Duration(seconds: 2)); // return ['Keyboard', 'Mouse', 'Monitor']; throw Exception('Gagal terhubung ke server'); }</pre> Tampilan: 
3.	Memulihkan kode dan menjalannkan ulang <pre>class ProductsNotifier extends AsyncNotifier<List<String>> { @override Future<List<String>> build() async { await Future.delayed(const Duration(seconds: 2)); return ['Keyboard', 'Mouse', 'Monitor']; } }</pre> Tampilan :



	
4.	Refleksikan: mengapa menampilkan ulang data lama (stale data) dengan indikator refresh kadang lebih baik daripada mengosongkan layar? Kapan pola itu penting? Answer: Menampilkan data lama (<i>stale data</i>) dengan indikator <i>refresh</i> menjaga kenyamanan antarmuka agar pengguna tidak kebingungan atau terganggu oleh layar yang tiba-tiba menjadi putih/kosong saat memuat pembaruan. Pola ini sangat penting diterapkan pada aplikasi yang membutuhkan proses penarikan data berulang—seperti <i>feed</i> media sosial, kotak masuk pesan, atau dasbor analitik—sehingga pengguna tetap memiliki konteks untuk dibaca sambil menunggu koneksi latar belakang selesai memuat data baru.

AI Challenge

Langkah	Keterangan
1.	Membuat stats page

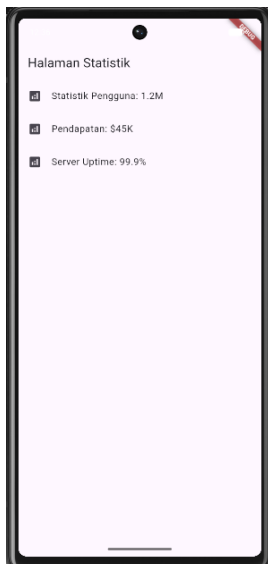


```
import 'dart:math';
import 'package:flutter/material.dart';
import 'package:flutter_riverpod/flutter_riverpod.dart';

// 1. Deklarasi Notifier menggunakan AsyncNotifier (Modern Riverpod API)
// Tidak menggunakan StateProvider atau StateNotifierProvider yang sudah usang.
class StatsNotifier extends AsyncNotifier<List<String>> {
  @override
  Future<List<String>> build() async {
    // Simulasi delay jaringan (loading) selama 2 detik
    await Future.delayed(const Duration(seconds: 2));

    // Simulasi kemungkinan gagal 30% menggunakan Random
    final isSuccess = Random().nextDouble() > 0.3;
    if (!isSuccess) {
      throw Exception('Gagal mengambil data statistik server.');
```

Tampilan :



2.

Testing



```
import 'package:flutter_test/flutter_test.dart';
import 'package:flutter_riverpod/flutter_riverpod.dart';
// Sesuaikan import ini dengan nama project Anda
// import 'package:week3_todo/pages/stats_page.dart';

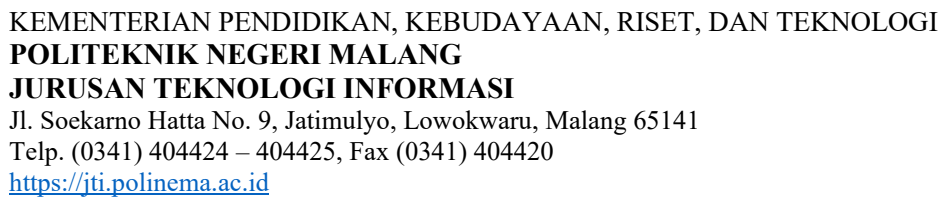
void main() {
  test('StatsNotifier memancarkan status loading lalu data atau error secara acak', () async {
    // 1. Buat ProviderContainer untuk menampung dan menguji state secara terisolasi
    final container = ProviderContainer();
    addTearDown(container.dispose);

    // 2. Verifikasi state awal adalah AsyncLoading sebelum delay selesai
    expect(
      container.read(statsProvider),
      const AsyncLoading<List<String>>(),
      reason: 'State pertama saat provider dibaca haruslah loading.',
    );

    // 3. Tunggu proses asinkron (delay 2 detik) selesai
    try {
      final data = await container.read(statsProvider.future);

      // 4a. Jika lolos dari peluang error 30% (sukses), pastikan state berupa AsyncData
      expect(container.read(statsProvider).value, equals(data));
      // Verifikasi data success memiliki 3 item
      expect(data.length, 3);
    } catch (e) {
      // 4b. Jika masuk ke peluang error 30% (gagal), pastikan state berupa AsyncError
      expect(container.read(statsProvider).hasError, isTrue);
      expect(
        container.read(statsProvider).error.toString(),
        contains('Gagal mengambil data statistik server.'),
      );
    }
  });
}
```

```
PS C:\Users\muham\Documents\Tugas Kuliah\Semester 5\Mobile\244107020112_mobile_course\week3_navigation> flutter test
00:04 +0: ... Kuliah\Semester 5\Mobile\244107020112_mobile_course\week3_navigation\test\stats_notifier_test.dart
test/stats_notifier_test.dart:14:22: Error: Undefined name 'statsProvider'.
  container.read(statsProvider),
                ^^^^^^^^^^^^^
test/stats_notifier_test.dart:21:41: Error: Undefined name 'statsProvider'.
  final data = await container.read(statsProvider.future);
                                ^^^^^^^^^^^^^
test/stats_notifier_test.dart:24:29: Error: Undefined name 'statsProvider'.
  expect(container.read(statsProvider).value, equals(data));
                        ^^^^^^^^^^^^^
test/stats_notifier_test.dart:30:29: Error: Undefined name 'statsProvider'.
  expect(container.read(statsProvider).hasError, isTrue);
                        ^^^^^^^^^^^^^
test/stats_notifier_test.dart:32:24: Error: Undefined name 'statsProvider'.
  container.read(statsProvider).error.toString(),
                ^^^^^^^^^^^^^
test/stats_notifier_test.dart:26:19: Error: The getter 'length' isn't defined for the type 'Object?'.
- 'Object' is from 'dart:core'.
Try correcting the name to the name of an existing getter, or defining a getter or field named 'length'.
  expect(data.length, 3);
                ^^^^^
00:09 +0 -1: loading C:\Users\muham\Documents\Tugas Kuliah\Semester 5\Mobile\244107020112_mobile_course\week3_navigation\test\stats_notifier_test.dart [E]
  Failed to load "C:\Users\muham\Documents\Tugas Kuliah\Semester 5\Mobile\244107020112_mobile_course\week3_navigation\test\stats_notifier_test.dart":
  Compilation failed for testPath=C:\Users\muham\Documents\Tugas Kuliah\Semester 5\Mobile\244107020112_mobile_course\week3_navigation\test\stats_notifier_test.dart: test/stats_notifier_test.dart:14:22: Error: Undefined name 'statsProvider'.
    container.read(statsProvider),
                    ^^^^^^^^^^^^^
test/stats_notifier_test.dart:21:41: Error: Undefined name 'statsProvider'.
  final data = await container.read(statsProvider.future);
                                ^^^^^^^^^^^^^
test/stats_notifier_test.dart:24:29: Error: Undefined name 'statsProvider'.
  expect(container.read(statsProvider).value, equals(data));
                        ^^^^^^^^^^^^^
```



Refactoring dan testing

Langkah	Keterangan
1.	Memisahkan TodoTile & Ekstrak Logika Filter



	<pre>import 'package:flutter/material.dart'; import 'package:flutter_riverpod/flutter_riverpod.dart'; class Todo { Todo(this.title, {this.done = false}); final String title; final bool done; Todo copyWith({String? title, bool? done}) => Todo(title ?? this.title, done: done ?? this.done); } class TodoListNotifier extends Notifier<List<Todo>> { @override List<Todo> build() => const []; void add(String title) => state = [...state, Todo(title)]; void toggle(int index) { final todos = [...state]; todos[index] = todos[index].copyWith(done: !todos[index].done); state = todos; } void remove(int index) => state = [...state]..removeAt(index); } final todoListProvider = NotifierProvider<TodoListNotifier, List<Todo>>(TodoListNotifier.new); final incompleteTodoProvider = Provider<List<Todo>>((ref) { final todos = ref.watch(todoListProvider); return todos.where((todo) => !todo.done).toList(); }); class TodoTile extends ConsumerWidget { final Todo todo; final int index; const TodoTile({super.key, required this.todo, required this.index}); @override Widget build(BuildContext context, WidgetRef ref) { return ListTile(leading: Checkbox(value: todo.done, onChanged: (_) => ref.read(todoListProvider.notifier).toggle(index),), title: Text(todo.title, style: TextStyle(decoration: todo.done ? TextDecoration.lineThrough : null,),), trailing: IconButton(</pre>
2.	<pre>) } class TodoPage extends ConsumerWidget { const TodoPage({super.key}); @override Widget build(BuildContext context, WidgetRef ref) { final todos = ref.watch(todoListProvider); return Scaffold(appBar: AppBar(title: const Text('ToDo Riverpod')), body: todos.isEmpty ? const Center(child: Text('Belum ada tugas')) : ListView.builder(itemCount: todos.length, itemBuilder: (context, index) { return TodoTile(todo: todos[index], index: index); },), floatingActionButton: FloatingActionButton(onPressed: () => _showAddDialog(context, ref), child: const Icon(Icons.add),),); } void _showAddDialog(BuildContext context, WidgetRef ref) { final controller = TextEditingController(); showDialog(context: context, builder: (context) => AlertDialog(title: const Text('Tugas baru'), content: TextField(controller: controller, autofocus: true), actions: [TextButton(onPressed: () => Navigator.pop(context), child: const Text('Batal'),), FilledButton(onPressed: () { if (controller.text.trim().isEmpty) { ref.read(todoListProvider.notifier).add(controller.text.trim()); } Navigator.pop(context); }, child: const Text('Tambah'),),],),); } }</pre>



	<pre>import 'package:flutter/material.dart'; import 'package:flutter_riverpod/flutter_riverpod.dart'; import 'package:go_router/go_router.dart'; import 'pages/todo_page.dart'; import 'pages/stats_page.dart'; void main() => runApp(const ProviderScope(child: MyApp())); // Widget kerangka utama yang memiliki NavigationBar[cite: 22] class MainScaffold extends StatelessWidget { final Widget child; const MainScaffold({super.key, required this.child}); @override Widget build(BuildContext context) { return Scaffold(body: child, bottomNavigationBar: NavigationBar(onDestinationSelected: (index) { if (index == 0) context.go('/'); if (index == 1) context.go('/stats'); }, destinations: const [NavigationDestination(icon: Icon(Icons.list), label: 'ToDo'), NavigationDestination(icon: Icon(Icons.analytics), label: 'Stats'),],),); } } final _router = GoRouter(initialLocation: '/', routes: [ShellRoute(builder: (context, state, child) => MainScaffold(child: child), routes: [GoRoute(path: '/', builder: (context, state) => const TodoPage()), GoRoute(path: '/stats', builder: (context, state) => const StatsPage()),],),],); class MyApp extends StatelessWidget { const MyApp({super.key}); @override Widget build(BuildContext context) { return MaterialApp.router(title: 'Week 3 - Refactoring', routerConfig: _router, theme: ThemeData(colorSchemeSeed: Colors.teal, useMaterial3: true),); } }</pre>
3.	Testing <pre>import 'package:flutter_test/flutter_test.dart'; import 'package:flutter_riverpod/flutter_riverpod.dart'; import 'package:week3_navigation/main.dart'; void main() { testWidgets('menambah tugas baru', (tester) async { await tester.pumpWidget(const ProviderScope(child: MyApp())); expect(find.text('Belum ada tugas'), findsOneWidget); await tester.tap(find.byIcon(Icons.add)); await tester.pumpAndSettle(); await tester.enterText(find.byType(Textfield), 'Kerjakan PR minggu 3'); await tester.tap(find.text('Tambah')); await tester.pump(); expect(find.text('Kerjakan PR minggu 3'), findsOneWidget); }); }</pre> Verifikasi:



```
25890E2..653f8ea main -> main
PS C:\Users\muham\Documents\Tugas Kuliah\Semester 5\Mobile\244107020112_mobile_course\week3_navigation> flutter analyze
Analyzing week3_navigation...

error - Undefined name 'statsProvider'. Try correcting the name to one that is defined, or defining the name -
test\stats_notifier_test.dart:14:22 - undefined_identifier
error - Undefined name 'statsProvider'. Try correcting the name to one that is defined, or defining the name -
test\stats_notifier_test.dart:21:41 - undefined_identifier
error - Undefined name 'statsProvider'. Try correcting the name to one that is defined, or defining the name -
test\stats_notifier_test.dart:24:29 - undefined_identifier
error - The property 'length' can't be unconditionally accessed because the receiver can be 'null'. Try making the acces-
conditional (using '?') or adding a null check to the target ('!') - test\stats_notifier_test.dart:26:19 -
unchecked_use_of_nullable_value
error - Undefined name 'statsProvider'. Try correcting the name to one that is defined, or defining the name -
test\stats_notifier_test.dart:30:29 - undefined_identifier
error - Undefined name 'statsProvider'. Try correcting the name to one that is defined, or defining the name -
test\stats_notifier_test.dart:32:24 - undefined_identifier
error - Undefined name 'Icons'. Try correcting the name to one that is defined, or defining the name -
test\widget_test.dart:10:34 - undefined_identifier
error - Undefined name 'TextField'. Try correcting the name to one that is defined, or defining the name -
test\widget_test.dart:13:40 - undefined_identifier

8 issues found. (ran in 47.3s)
PS C:\Users\muham\Documents\Tugas Kuliah\Semester 5\Mobile\244107020112_mobile_course\week3_navigation> flutter analyze
Analyzing week3_navigation...

warning - The '!' will have no effect because the receiver can't be null. Try removing the '!' operator -
test\stats_notifier_test.dart:22:18 - unnecessary_non_null_assertion

1 issue found. (ran in 25.4s)
PS C:\Users\muham\Documents\Tugas Kuliah\Semester 5\Mobile\244107020112_mobile_course\week3_navigation> flutter analyze
Analyzing week3_navigation...
No issues found! (ran in 23.1s)

PS C:\Users\muham\Documents\Tugas Kuliah\Semester 5\Mobile\244107020112_mobile_course\week3_navigation> flutter test
composing: TextRange(start: -1, end: -1))), focusNode: FocusNode#3305b, autocorrect: true,
debugLabel: ((englishLike bodyLarge 2021)).merge((blackMountainView
bodyLarge).apply)).merge(unknown), inherit: false, color: Color(alpha: 1.0000, red: 0.0863, green:
0.1137, blue: 0.1098, colorSpace: ColorSpace.sRGB), family: Roboto, size: 16.0, weight: 400,
letterSpacing: 0.5, baseline: alphabetic, height: 1.5x, leadingDistribution: even, decoration:
Color(alpha: 1.0000, red: 0.0863, green: 0.1137, blue: 0.1098, colorSpace: ColorSpace.sRGB)
TextDecoration.none, textAlign: start, autofocus: true, keyboardType: TextInputType(name:
TextInputType.text, signed: null, decimal: null), autofillHints: [], spellCheckConfiguration:
SpellCheckConfiguration(disabled, service: null, text style: null, toolbar builder: null),
dependencies: [Directionality, MediaQuery, _EffectiveTickerMode, _FocusInheritedScope, _ViewScope],
state: EditableTextState#83df5(tickers: tracking 1 ticker)),
]>
Which: is too many

When the exception was thrown, this was the stack:
#4 main.<anonymous closure> (file:///C:/Users/muham/Documents/Tugas%20Kuliah/Semester%205/Mobile/244107020112_mobile_course/we
ek3_navigation/test/widget_test.dart:20:5)
<asynchronous suspension>
#5 testWidgets.<anonymous closure>.<anonymous closure> (package:flutter_test/src/widget_tester.dart:192:15)
<asynchronous suspension>
#6 TestWidgetsFlutterBinding._runTestBody (package:flutter_test/src/binding.dart:1953:5)
<asynchronous suspension>
<asynchronous suspension>
(elided one frame from package:stack_trace)

This was caught by the test expectation on the following line:
file:///C:/Users/muham/Documents/Tugas%20Kuliah/Semester%205/Mobile/244107020112_mobile_course/week3_navigation/test/widget_test.
dart line 20
The test description was:
menambah tugas baru

00:14 +1 -1: C:/Users/muham/Documents/Tugas Kuliah/Semester 5/Mobile/244107020112_mobile_course/week3_navigation/test/widget_test.d
art: menambah tugas baru [E]
Test failed. See exception logs above.
The test description was: menambah tugas baru

To run this test again: C:\flutter\flutter\bin\cache\dart-sdk\bin\dart.exe test C:/Users/muham/Documents/Tugas Kuliah/Semester 5/Mo
bile/244107020112_mobile_course/week3_navigation/test/widget_test.dart -p vm --plain-name "menambah tugas baru"
00:14 +1 -1: Some tests failed.
```

Refleksi

1. Kapan setState masih cukup, dan kapan state harus naik ke Riverpod?

Answer :



setState cukup untuk *state* lokal yang hanya berlaku di satu *widget* (seperti status *checkbox* atau animasi sementara). *State* harus naik ke Riverpod jika datanya perlu diakses oleh banyak halaman, logika bisnisnya kompleks, atau terhubung dengan data dari API/Database.

2. Apa perbedaan context.go dan context.push, dan kapan masing-masing tepat digunakan?

Answer :

context.go menimpa rute saat ini secara absolut (*replace*), sehingga cocok digunakan untuk navigasi menu utama. Sebaliknya, context.push menumpuk halaman baru di atas rute sebelumnya (*stack*), sehingga ideal untuk membuka halaman detail yang membutuhkan tombol "Kembali".

3. Bagaimana AsyncValue mencegah bug dibanding tiga boolean terpisah?

Answer:

Tiga boolean terpisah rawan saling tumpang tindih (misalnya status *loading* dan *error* aktif bersamaan). AsyncValue mencegah *bug* tersebut dengan menerapkan kondisi yang ketat secara eksklusif (hanya bisa *loading*, *error*, atau *data* pada satu waktu) dan memaksa *developer* menangani ketiga tampilan UI-nya.

4. Bagian mana dari hasil AI yang Anda perbaiki, dan mengapa?

Answer :

Saya memperbaiki *path import* file yang tidak sesuai dengan direktori lokal agar tidak *error*, serta menghapus operator asersi ! yang berlebihan pada *unit test*. Penghapusan ini dilakukan karena fitur *null-safety* Dart telah mendeteksi variabel tersebut aman, sehingga peringatan pada flutter analyze bisa hilang.